Building Reliability Across Cloud Boundaries

Lessons from the Multicloud Journey

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The New Reality

1,529%

Oracle Multicloud Revenue Growth

Enterprises are going multicloud.

- They're not asking IF they should use multiple clouds
- They're asking HOW to do it reliably
- They expect the same SLAs everywhere

"Your customers chose their cloud. You must meet them where they are."

Multicloud Patterns

A spectrum defined by coupling depth — and the latency tier it enables

← loosely coupled

tightly coupled →

1

Containerized Deployments

Cloud-agnostic portability

— Same container image runs on any cloud provider

— Kubernetes / managed K8s on Cloud A, B, C

— OCI, Helm, IaC, service mesh, CI/CD pipelines

Value: workload portability, avoid lock-in

No cross-cloud I/O

2

App Layer Integration

App layer integration

— Each SaaS stays sovereign on its home cloud

— Private interconnect carries workflow & events

— Data replication, event streaming, identity federation

Value: business agility without re-architecture

Application-layer

3

Network Layer Integration

Network-coupled tiers

— App tier on Cloud A/B; database tier on OCI

— Dedicated private WAN via 3rd-party fabric

— Best-of-breed DB engine, independent scaling

Value: leverage existing app + superior DB stack

2 – 10 ms (WAN)

4

Co-existing Cloud

Physical colocation

— Oracle DB runs in an OCI child site inside AWS / Azure / GCP datacenter

— Unified console, billing, and identity (SSO)

— No data egress fees, feature + price parity

Value: AI / real-time OLTP at full cloud scale

< 1 ms (µs-class)

· Colocation is the only path to microsecond latency · Colocated Clouds requires a commercial partnership between providers

The Multicloud Provider Challenge

You Own the Experience

But not the infrastructure beneath it

Different Integration Models

Every cloud partner has unique requirements

Your SLAs Depend on Theirs

SLAs you didn't negotiate, can't control

Incidents Cross Boundaries

Different APIs, failure modes, escalation paths

Designing for Multicloud Service Delivery

01

The Reusable Framework Approach

Build once, deploy everywhere—without lowest-common-denominator reliability. Abstract the differences, but don't ignore them.

02

Abstraction Without Ignorance

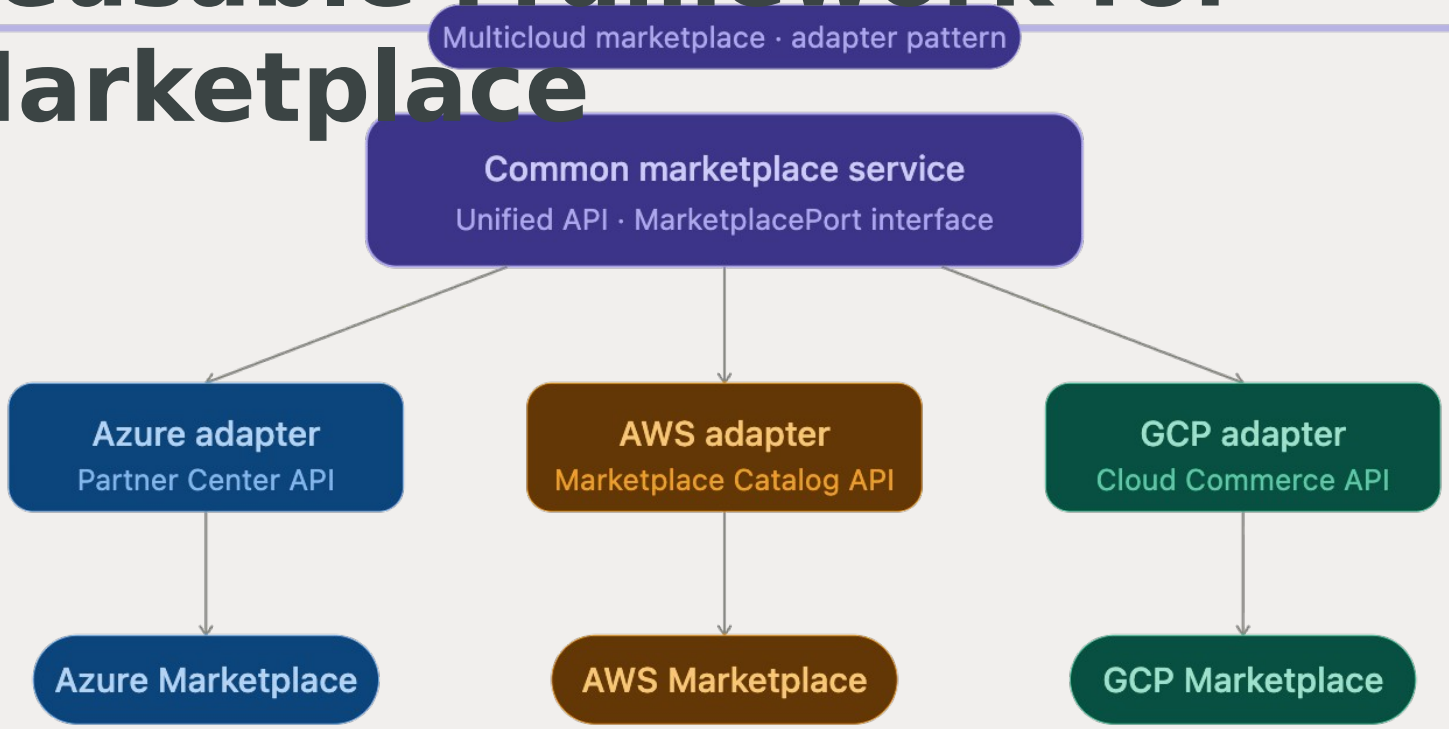
Hide cloud differences from your service code while remaining deeply aware of them in your SRE practices and runbooks.

03

Cloud-Native vs Cloud-Agnostic

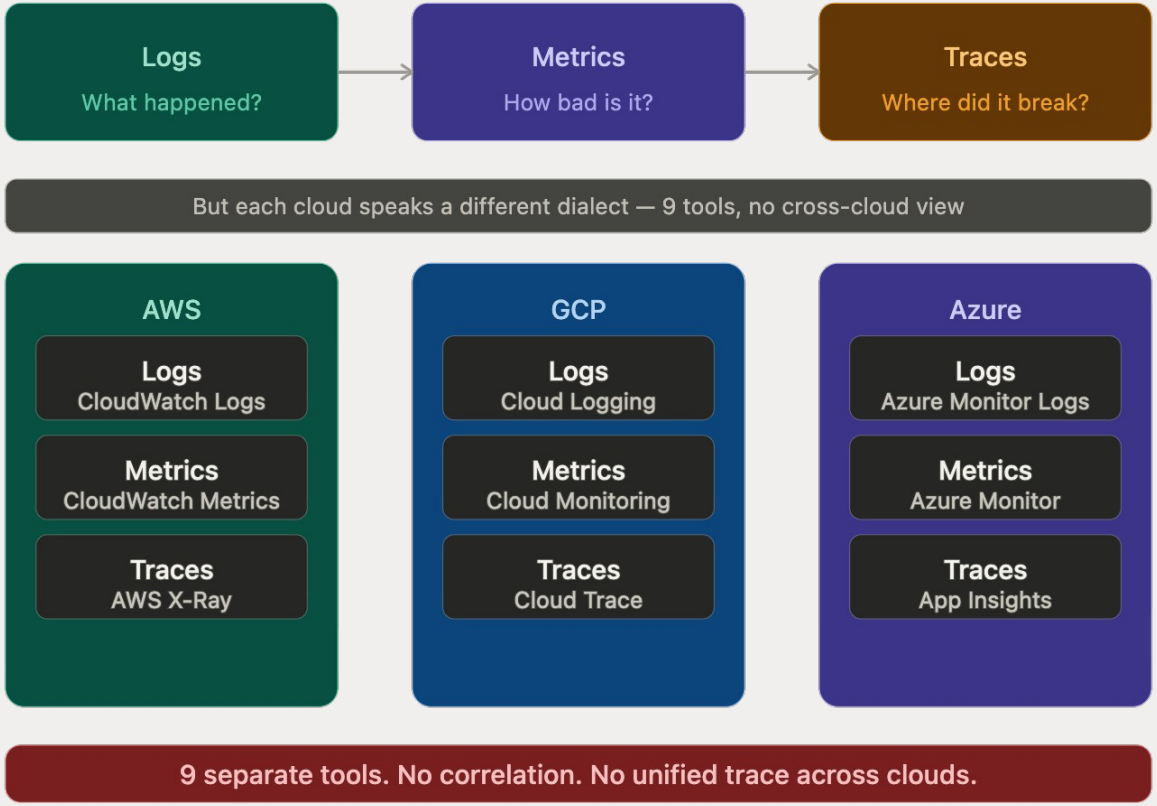
Know when to embrace a cloud's native services (marketplace, observability, IAM) and when to bring your own (core business logic).

Reusable Framework for Marketplace

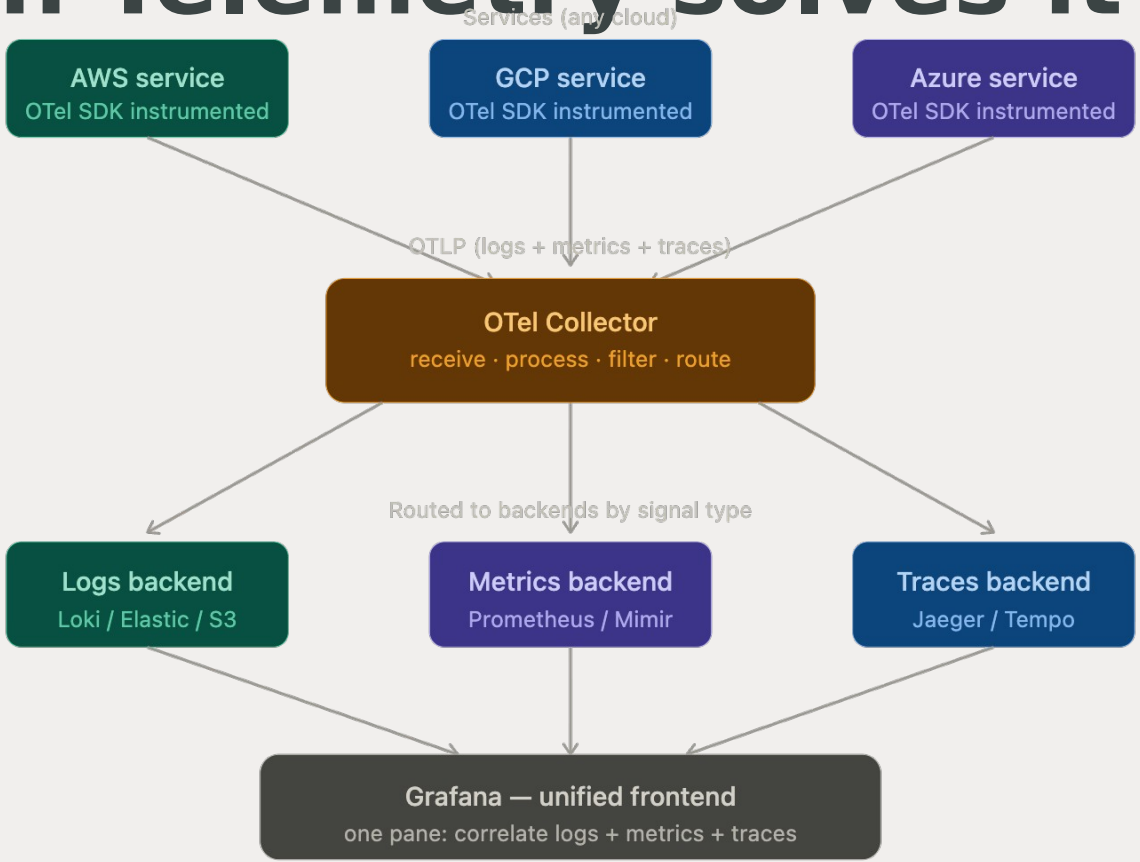


The Observability Problem

The 3 pillars — and why multicloud breaks them



Open Telemetry solves it



Operating Across Cloud Boundaries

Capacity Planning

Region rollouts across GCP, Azure, AWS

Variable run-rate per cloud partner

Check partner capacity before promising customers

Deployment & Release

Canary deployments in multicloud

Coordinated maintenance windows

Cloud-aware rollback triggers

Incident Response

Runbooks that branch by cloud

Escalation paths include partners

Chaos engineering respects partner boundaries

Scaling the Organization

No single engineer can master four clouds. How do you structure expertise?

Knowledge Architecture

- Cloud champions (not silos)
- Cross-training rotations
- Shared incident reviews

Documentation at Scale

- Runbooks with cloud branches
- Living architecture docs
- Decision logs for why

Partner Relationships

- Regular sync meetings
- Shared Slack channels
- Joint incident reviews

Lessons Learned

- ✓ Build observability BEFORE architecture—retrofitting across four clouds is exponential
- ✓ Standardize what you can, adapt what you must, document the differences obsessively
- ✓ Reliability is a partnership—with cloud providers, customers, and your distributed teams
- ✓ Invest in partner relationships before you need them in a crisis

The Road Ahead

1

AI-Powered Operations

Cross-cloud anomaly detection, predictive scaling, digital twin and world model for causal analysis

2

Edge & Sovereign Clouds

New boundaries to cross, new compliance requirements, data residency

3

Post-Quantum Cryptography

Securing data across clouds for the quantum era- migration starts now

4

Operational Excellence as Differentiator

As more providers go multicloud, reliability becomes the competitive edge

Key Takeaways

1. Delivering services across clouds ≠ consuming multiple clouds
2. Observability strategy before architecture
3. Reliability is a partnership, not a solo effort
4. Standardize, adapt, document obsessively
5. The SRE discipline must evolve for multicloud

Thank You!

Questions?

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